

Da Nang, January 20th 2026

CAPSTONE PROJECT PROPOSAL

1. Student Information:

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Supervisor: Dr. Nguyen Quang Vu Co-Supervisor: Assoc. Prof. Dr. Sunarin Chanta	

2. Project Title: Developing a Multi-Objective Evolutionary Algorithm for Disaster Relief Hub Network Design.

3. Description:

Overview: This project focuses on optimization of Humanitarian Logistics networks for disaster response. The problem is framed as a Multi-objective Incomplete Hub Location Network Design Problem (MO-IHLNDP).

Objectives: To develop and enhance an evolutionary algorithm (based on NSGA-II) to simultaneously optimize three objectives: (1) Minimize total operational costs, (2) Minimize minimax travel time for emergency response, and (3) Maximize relief coverage.

Expected Results:

1. A mathematical formulation for the Humanitarian MO-IHLNDP.
2. An efficient algorithm implementation tested on the standard TR Dataset (Turkish Network - 81 nodes).
3. A Graduation Thesis report and a scientific paper submitted to the CITA 2026 conference. (<https://cita-conf.org/>)

4. Research Tasks:

1. Literature Review: Review prior works on Humanitarian Logistics, Hub Location Problems (HLP), and existing solution methods.
2. Problem Formulation: Define objective functions (Z1: Cost, Z2: Time, Z3: Coverage), decision variables, and specific constraints for incomplete networks.
3. Algorithm Development: Develop and improve an evolutionary algorithm (building on previous internship work) to handle minimax objectives and disaster-specific constraints.
4. Experiments & Evaluation: Conduct simulations on the TR dataset, analyze Pareto optimal solutions, and perform comparisons on the benchmark dataset.

5. **Writing & Publication:** Finalize the thesis report and draft a research paper for CITA 2026 co-authored with supervisors.

5. Implementation Plan:

Time	Job to be done
Week 1 - 2	Literature review; TR Dataset acquisition and preprocessing.
Week 3 - 4	Mathematical modeling (problem formulation); Algorithm high-level design.
Week 5 - 8	Algorithm implementation and tuning in C++ / Python; Preliminary experiments.
Week 9 - 10	Analysis of experimental results; Comparison with prior works; Drafting the CITA 2026 paper.
Week 11 - 12	Finalizing the CITA paper; Writing the Graduation Thesis; Preparing defense presentation.

January, 2026

Supervisor

January, 2026

Co-Supervisor

January, 20th 2026

Student

Nguyễn Quang Vũ

Sunarin Chanta

Lê Tăng Phú Quý